Data Analytics Report

Week 4 – Final Presentation & Insights Report

AI Powered Data Analysis Remote Internship

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Team name: Team 6

Dataset: Student Participation Dataset

Contributions

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Executive Summary – Genevieve Ah Shene & Usman Bin Yousuf

Historical Analysis – Busayo Salau

Seasonal Trend Analysis & Conclusion – Usman Bin Yousuf

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Forecast Results & Model Evaluation - Yonnie Ntsubane & Sudheer Muthyala

Strategic Recommendations – Chiamaka Chinaza

Business Impact Assessment – Rejoice Daniel

Introduction

What was the purpose of the dataset?

Analyzing historical learner application data from RIT Student Participation Dataset.

3 questions while analyzing:

1. Where is demand concentrated?

2. When does demand peak?

3. What should program manager do about it?

What was team approach?

- Data collection & cleaning
- EDA
- Trend Analysis across countries, months, quarters, and opportunity categories.
- Predictive modeling to estimate future application demand.
- Visualizations and actional insights to support data-driven decisions.
- What models were used for predictive modeling?
- Random Forest Model: Primary Forecasting model
- XGBoost – Independent model for cross-checking forecasts.

Executive Summary

- Demand peaks in Q1, with secondary peak in Q3.
- Demand is concentrated: Internship and Courses make up about 87% of applications.
- Application period: October 2022 to Apr 2024.

- Top university: Saint Louis University
- Total application records: 8553.
- Top category share: 63.4% Internships
- The model confirms the Q1 peaks
- Conversion is the bottleneck: 42% of applications rejected, 9% started.

Use of AI

Tools used: Claude AI (Sentence Structure and KPI research)

Prompt:

- Provide the KPIs for the dataset provided based on the reports from weeks 1 to 3.
- Create KPI dashboard visuals for an executive summary based on this information.

Explain how the two models were used for predictive modeling using the reports

Historical Analysis

Learner Signup: This year and quarterly signups for 2023:

- Q1: 500
- Q2: 1435
- Q3: 2503
- Q3: 1270

August showed the highest monthly signup of 1133 while Q1 2024 had 2739 signups with January having 1547 signups.

Status Description: Out of 8553 learners, the percentages of rejected, team allocated, started, dropped out, waitlisted, applied, withdraw and rewards awards were:

- Rejected: 41.72%
- Team Allocated: 38.30%
- Started: 8.93%
- Dropped Out: 7.20%
- Waitlisted: 1.27%
- Applied: 1.23%
- Withdrew: 1.10%
- Rewards Award: 0.34%

Country: United States of America has the highest number of learners (3976) followed by India with (2832) and Rwanda accounted for the lowest number of participants (28).

Thirty new features were created in addition to the 16 making a total of 46 features.

Among the opportunity categories: Internship accounted for the highest number of participants (63.4%), while engagement had the lowest number of participants (1.5%). Others were course, event and competition accounting for 23.8%, 6.4% and 5% respectively.

Predictive modeling was done using two models: Random Forest and XGBoost and performances were based on accuracy, precision, recall, F1-score and AUC score. Using 5-fold CV and top 3 features led to accuracy of 94.9% for Random Forest and 93.3% for XGBoost

Seasonal Trend Analysis

Application activity shows a fluctuating seasonal pattern rather than a consistent long-term increase or decline. Activity increased strongly through 2023, reaching high levels in August, followed by declines during September–November.

Participation then rebounded sharply in December and January, with January 2024 recording the highest monthly volume. Internship participation shows the strongest seasonal pattern, peaking at around 1,400 learners in January with a secondary peak in August. Other types of opportunity peak at different times, with courses highest in June and events and competitions in February

Overall, January–February and August appear to be important periods for learner engagement, although the exact peak varies by opportunity type.

Forecasting Result

The Random Forest (RF) model was used to forecast demand for opportunity categories and quarterly application counts. The forecasting analysis involved comparing model predictions to observed actual values to determine how closely the model reflected participation patterns within the dataset. The results show the Random Forest can capture the overall distribution of opportunity demand and can clearly distinguish the application volume by quarter.

Opportunity Category Forecast

The Random Forest model predicted internships to be the dominant opportunity category with 5306 predicted applications against 5037 actual records. Courses were predicted for 2011 applications compared to 2013 actual records. Events, Competitions and Engagement were also predicted close to their observed values.

Opportunity Category	Actual	RF predicted	Difference
Internship	5307	5306	+269
Course	2013	2011	-2
Event	537	536	-1
Competition	420	425	+5
Engagement	128	127	-1

The model kept the same overall rank found in the data, with Internships responsible for about 63% of predicted opportunity demand and Courses for about 24%. Together, these two groups account for some 87% of the predicted demand, indicating that career-oriented opportunities should be a major consideration in the planning of future programs and resource allocation.

Quarterly Application Forecast

The volume of applications for the four quarters was also estimated using the Random Forest model. The model predicted the highest demand for applications in Q1 with 4924 predicted applications versus actual applications of 4871. This is the most accurate RF prediction for the peak quarter and confirms the finding that Q1 is the strongest period of application demand.

Quarter	Actual Applications	RF Predicted	Difference
Q1	4871	4924	+53
Q2	605	530	-75
Q3	1690	1744	+54
Q4	1239	1207	-32

Forecast Reliability and Limitations

The forecasting results provide useful directional rights, particularly on the high-demand opportunity categories and seasonal application patterns. However, the results should be interpreted with caution. In the analysis of Week 3, the possibility of data leakage was discussed especially in the prediction task of the opportunity category where information regarding the outcome could have influenced the model's performance. The report therefore recommends that the predictions be further validated using leakage-free, time-based testing before they are used for operational decision making.

Overall, the Random Forest results indicate that Q1 is probably going to continue to be the busiest time for applications and internships are probably going to continue to be the most

sought-after opportunity type. These results can help with resource allocation and program planning, but model validation is advised to increase confidence in precise forecast values.

Opportunity Category Ranking

The historical analysis ranks opportunity categories based on their application volume:

Rank Opportunity Category Applications

1. Internship: 5,419
2. Course: 2,034
3. Event: 545
4. Competition: 425
5. Engagement: 130

Key Findings:

Internships represent the highest application volume, followed by courses.

Together, Internships and courses account for approximately 87.2% of the 8553 historical records.

Events, Competitions and engagement opportunities receive considerably fewer applications.

Machine Learning Forecast Findings

The week 3 analysis uses Random Forest and XGBoost to predict opportunity category distribution. Both models produce the same ranking: Internship, Course, Event, Competition and Engagement.

The week 3 report records accuracy scores of 99.64% for Random Forest and 93.10% for XGBoost. However, the report also identifies potential data leakage concern, so these results should be validated before being used for operational planning.

Strategic Recommendations

1. **Prioritize capacity planning**

Align staffing and resources for internships and courses with expected demand.

2. **Review of lower-volume categories**

Review lower volume categorie by investigating awareness, eligibility requirements and participation barriers before deciding whether to change their scale.

3. **Validate ML predictions**

Compare future predictions against actual application volumes before relying on them for resource allocation.

4. **Maintain category diversity**

Consider program objectives and participant outcomes alongside application volume.

Business Impact and Measurement

These recommendations could improve resource allocation, program planning and participation, their impact should be measured using:

- Category wise application growth.
- Application- to start -date.
- Programming completion and dropout rates.

Forecast error between predicted and actual application records.

Program managers should establish a baseline and review these indicators monthly or after each program cycle.

Strategic Recommendations

1. Strategic Outreach plan

Target Career Focused Demand: Prioritize marketing for Internships and Courses as together they represent 87% of all learner demand.

Segment Engagement: Maintain secondary outreach channels for Events, Competition, and Engagement opportunities (1.5% share) to avoid over-allocating marketing budgets where demand is low.

Timing-driven outreach: Application timing is the single strongest predictor of participation (41 – 42% importance), launch early awareness campaigns 60-90+ days prior to program start dates to maximize acceptance and signup rates.

2. Resource allocation strategy

Front-load capital & support in Q1: Focus operational staffing, server capacity, and reviewer bandwidth in Q1, which represents the peak demand season.

Scale down operations for Q2-Q4: Shift resources to lower-demand quarters to focus on program execution, curriculum development, and data cleaning rather than heavy recruitment.

Capacity planning by category: Allocate mentor pools and host organization partnerships primarily to Internship programs.

3. Promotional & Scheduling Window

Early-bird lead times: Leverage the finding that learners applying far in advance show up to a 99% acceptance rate. Promote opportunities early rather than relying on last-minute application drives.

Align with signup windows: Capitalize on learner signup quarter – the top feature driving quarterly demand by automatically serving tailored course/internship promotions immediately after a learner creates an account.

Program duration transparency: Clear visibility into program duration increases conversion, structure promotional materials around clear start/end dates and overall program length.

4. Measuring Impact and Future Learning Opportunities

Audit for data leakage: Implement time-based cross-validation after removing target-leakage features such as Status Description. Retain baseline testing across clean datasets.

Monitor surge forecast errors: Refine time-series modeling to address underestimations during sudden surges.

Track conversion & retention metrics: Measure key performance indicators (KPIs) by opportunity category, including application-to-acceptance ratios across predicted probability bands.

What programs managers should do next & why it matters

Priority Action for Program Managers	**Implementation Steps**	**Why it matters**
1. Fixed Data leakage in models	Work with the data analytics team to strip out Status Description and outcome-based features from the training pipeline, then re-evaluate RF and XGBoost models.	**Model Integrity:** Current ~99.64% - 100% accuracy metrics are misleading. Decisions made on flawed models risk misallocating operational funds and staffing.
2. Preparing Operations	Reallocate administrative staff, platform infrastructure and reviewer capacity to handle	**Operational Efficiency:** Prevents system bottlenecks, delayed

for Q1 surge	(58% - 60%) the annual volume in Q1.	applications reviews, and poor candidate experiences during peak demand
3. Rebalance Program Portfolio	Shift resource emphasis towards Internships and Courses, while offering Competition and Events on a leaner, automated scale.	**Resource Optimization:** Ensures 87% of organizational resources align with where 87% of learner demands.
4. Automate Post-Signup Nurture Workflows	Set up automated email/platform triggers delivering Internship/Course options immediately after learner signup	**Conversion Rates**: Signup timing is the number 1 indicator of learner application quarterly demand (59%-66%) importance

Business Impact of Key Findings

Resource allocation: Internships (63.4%) and Courses (23.8%) make up about 87% of demand, so resources should follow them.

Demand planning: Q1 holds about 58% of applications, so capacity should be ready before it.

Target communication: Application timing is the top predicator (41-42%), which makes the early window the best time to reach learners.

Retention: 612 Internship dropouts and 3568 rejections are clear targets. Illustrative: 10%-20% fewer dropouts keep about 61 – 122 learners.

Limitations and risks: Possible data leakage, the missed Q2 2024 surge (about 2000 forecast vs 4696 actual) and 105 data anomalies. Participation is concentrated, with Saint Louis University at about 51% of learners.

Mitigation: Retrain the models without outcome fields, treat forecasts as ranges, widen recruitment, and add data validation.

Key Takeaways:

These findings can help RIT improve resource applications, plan for demand, increase engagement, and make more informed program decisions.

Recommended Actions & Expected Impact

What should RIT do differently?

Priority	Action	Evidence	Expected Impact
1	Retrain and validate the model without leakage	Near-perfect accuracy, 40-48% backtest error	Reliable decisions
2	Early nudges and check-ins	Timing is the top predicator; 612 dropouts	Fewer dropouts, higher completion.
3	Prepare for Q1	58% of applications in Q1.	Better capacity and response times.
4	Prioritize Internships and Courses	87% of demand	Support where demand is.
5	Tailor outreach by learner profile (new vs returning, Internship vs Course)	Courses start rate 37.6%	Scalable engagement

Measurement

- Application Rate
- Participation Rate
- Conversion Rate
- Completion Rate
- Q1 Demand
- Prediction Accuracy

Conclusion

Key Takeaways:

Demand Concentration: Internships and courses account for approximately 87.2% of participation, making them the primary areas for resource allocation.

Seasonal Patterns: Applications activity fluctuates throughout the year, with notable peaks in Q1 and August, highlighting the importance of timely outreach and capacity planning.

Data-driven decisions: Random Forest and XGBoosr provided a foundational for forecasting demand, but further validation is required to address potential data leakage and forecast errors.

Moving Forward

RIT can use these insights to improve resource planning, target learner outreach, strengthen retention, and make more informed program decisions. Reliable data and validate forecasts will be essential to achieving these improvements.